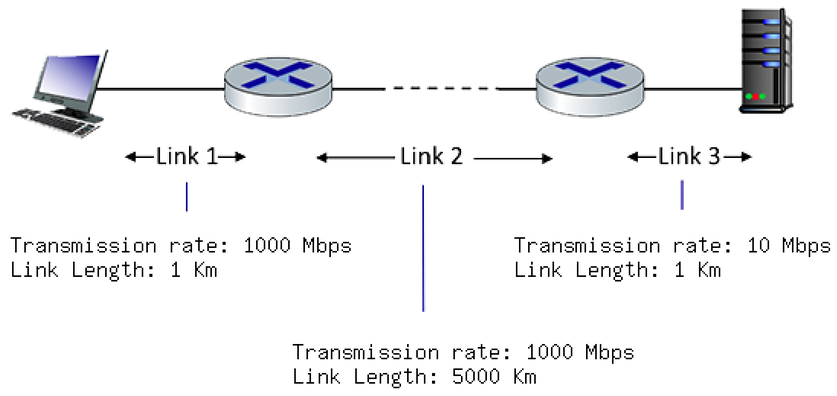
INTERACTIVE END-OF-CHAPTER EXERCISES

COMPUTING END-END DELAY (TRANSMISSION AND PROPAGATION DELAY)

Consider the figure below, with three links, each with the specified transmission rate and link length.



Assume the length of a packet is 8000 bits. The speed of light propagation delay on each link is 3x10^8 m/sec

Round your answer to two decimals after leading zeros

QUESTION LIST

1. What is the transmission delay of link 1?
2. What is the propogation delay of link 1?
3. What is the total delay of link 1?
4. What is the transmission delay of link 2?
5. What is the propogation delay of link 2?
6. What is the total delay of link 2?
7. What is the transmission delay of link 3?
8. What is the propogation delay of link 3?
9. What is the total delay of link 3?
10. What is the total delay?

SOLUTION

Link 1 transmission delay = L/R = 8000 bits / 1000 Mbps = 8.00E-6 seconds

Link 1 propagation delay = d/s = ()1 Km) * 1000 / 3*10^8 m/sec = 3.33E-6 seconds

Link 1 total delay = d_t + d_p = 8.00E-6 seconds + 3.33E-6 seconds = 1.13E-5 seconds

Link 2 transmission delay = L/R = 8000 bits / 1000 Mbps = 8.00E-6 seconds

Link 2 propagation delay = d/s = ()5000 Km) * 1000 / 3*10^8 m/sec = 0.017 seconds

Link 2 total delay = d_t + d_p = 8.00E-6 seconds + 0.017 seconds = 0.017 seconds

Link 3 transmission delay = L/R = 8000 bits / 10 Mbps = 0.0008 seconds

Link 3 propagation delay = d/s = ()1 Km) * 1000 / 3*10^8 m/sec = 3.33E-6 seconds

Link 3 total delay = d_t + d_p = 0.0008 seconds + 3.33E-6 seconds = 0.0008 seconds

The total delay = d_L1 + d_L2 + d_L3 = 1.13E-5 seconds + 0.017 seconds + 0.0008 seconds = 0.017 seconds

QUESTION 1 OF 10

What is the transmission delay of link 1?

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

Copyright © 2010-2025 J.F. Kurose, K.W. Ross
Comments welcome and appreciated: kurose@cs.umass.edu